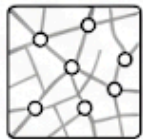


1. DATA INPUTS AND PREPROCESSING



Street Network and Existing Infrastructure (GIS data)



Level of Traffic Stress (LTS) Assessment



Bike-share OD Demand (Trip Records)



Cost Parameters and Budget Limits (\$)

2. MULTI-OBJECTIVE OPTIMIZATION MODEL

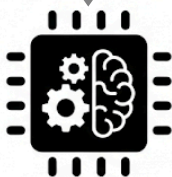
Obj. 1:
Maximize
Demand
Coverage



Obj. 2:
Minimize
LTS level
for users



Obj. 3:
Minimize
Upgrade
costs



Decision Variables: Links, stations, and number of docks



Constraint: Network connectivity and station capacity

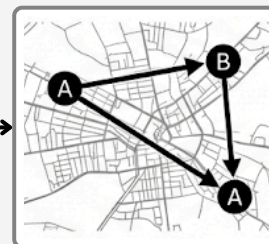
3. OPTIMAL DECISION RESULTS



Output 1:
New Bike-share
Stations
Locations



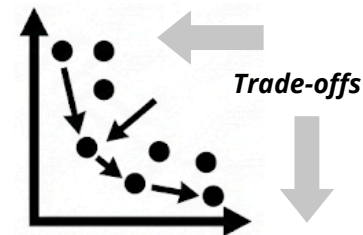
Output 2:
New Bicycle
Infrastructure
Links



Output 3:
Coordinated
Low-Stress
Network
Plan

4. EVALUATION AND INSIGHTS

Pareto Front Analysis
(Trade-off: Coverage vs.
Stress vs. Cost)



Sensitivity Analysis
(Coverage vs. Stress vs. Cost)

